

FPGA Development Solution

Vendor	Development Solution	Applications
Altera	Stratix IV GX FPGA Development Kit Cyclone V GX FPGA Development Kit Arria II GX FPGA Development Kit	Communication tester equipment production needs, ATE production and general-purpose test instrument and equipment production
Xilinx	Xilinx Virtex-7 FPGA VC709 Connectivity Kit	Ideal for designing high-end, chassis-based communications or video broadcast test equipment
	Spartan-3E Starter Kit	Spartan-3 FPGA features are ideal for applications including network analysers, protocol analysers and test equipment
National Instruments	NI PXIe-5644R/5645R module	Real-time signal processing and control for RF applications
Atmel	AT6000 FPGA Integrated Development System	Test equipment industrial control instrumentation special-purpose computers connectors
Microsemi	IGL002 FPGA Evaluation Kit	Enables evaluation and prototyping and to perform power measurements on low-power IGL002 FPGA
Extreme Engineering Solutions (X-ES)	XPedite2300 FPGA Development Kit	The XPedite2300 is well suited to data streaming applications that require real-time signal processing, such as video surveillance, signals intelligence, and infrared threat detection. It supports the Virtex-6 LX130T, LX195T, LX240T, LX365T, SX315T and SX475T FPGAs
Tekmicro	QuiXilica-V6 VME/VXS FPGA board	The QuXilica-V6 VME/VXS is compatible with legacy VME systems as well as newer ANSI/VITA 41 VXS-based systems for both laboratory and deployed/rugged applications
TEWS Technologies	TAMC651	Designed for industrial, COTS, and transportation applications, where specialised I/O or long-term availability is required, the TAMC651 provides a number of advantages including a customisable interface for unique applications and an FPGA-based design to extend product lifecycle

Stephen Hire, general manager, Aeroflex Systems Pvt Ltd, explains, "Product designers need test tools to be available at the same time as the products that they are developing, and that require significant investment by test equipment manufacturers. Taking our FPGA-based TM500 test mobile as an example, it is used by virtually every single base station manufacturer worldwide as a test peer because it was available long before commercial mobiles are."

Nate Srinath adds, "Since FPGAs are highly configurable, newer test cases and functionality can be embedded in a short period of time to allow the equipment manufacturers to reach market on a timely basis."

In-field firmware upgrades. Flex-

ibility and protection of investment are the two big advantages of in-field firmware upgrades enabled by FPGA-based equipment. "Technology, especially in the wireless industry, is constantly evolving, and different markets often have slightly varying standards or requirements, for example different spectrum allocations. When designers and manufacturers invest in new test equipment, they need tools that can not only cater to the variations across markets but also have the ability to be upgraded via software as standards and technology evolve. FPGAs are a major help in achieving these goals," shares Stephen Hire.

In-field firmware upgrades are a must for most modern-day embedded systems with communication

capabilities. Such systems are usually deployed remotely (like a microwave repeater) and need firmware updates to be deployed over the air or the provided network.

"It is unacceptable to expect the end customer to bring back the remote system to the factory for firmware upgrades. Customers can face substantial losses when firmware upgrades are not performed remotely. The system has to be brought down and back to the factory. During this time, the system is not operational and the cost of removal, transport and reinstallation adds to the losses due to the system downtime. It is strongly recommended that all embedded systems (wired or wireless) with communication capabilities should have in-field upgrades. This would add tremendous flexibility to the end customer to not only save costs, but provide the ability to change the system configuration remotely," adds Nate Srinath.

At the end of the day, is it worth it?

Whether to use FPGA in test depends on the feature and functionality requirement on the end users' test and measurement equipment.

Lavesh explains, "While on one hand, FPGAs in test are good for a system with multichannel parallel functionality and narrow bandwidth. FPGA-based test can also keep up with future modifications that might be necessary. On the other hand, a system with just one or two channels and of higher bandwidth (GHz), the cost of FPGA and thus the cost of the system for the end user can quickly exceed and it may not be worthwhile as compared to ASIC."

The fact that there exists a learning curve is in itself might also be a dampening factor for anyone looking to get into designing FPGA-based tools. However, Hanneke Krekels, test, measurement and emulation segment owner at Xilinx (an FPGA supplier for National Instruments), says, "Like with any new technology implementation, there can be a learning curve getting